13 Cultural Heritage Impact

13.1 Introduction

13.1.1.1 This Section of the report forms part of a Cultural Heritage Impact Assessment (CHIA). The CHIA is in accord with Appendix K of the EIA Study Brief No. ESB-246/2012 and presents the results of an assessment of potential impacts on cultural heritage resources such as terrestrial archaeological resources and historic buildings in the PDA as a result of construction and operational activities. Mitigation measures required to ameliorate the potential impacts to acceptable levels have been recommended, where appropriate. A full bibliography is presented of this report.

13.2 Environmental Legislation, Standards and Guidelines

13.2.1.1 The relevant legislation, standards and guidelines applicable to the present study for the assessment of cultural heritage impact and marine archaeology include:

- Antiquities and Monument Ordinance (Cap 53);
- Annex 10 of the EIAO-TM, Criteria for Evaluating Visual and Landscape Impact, and Impact on Sites of Cultural Heritage;
- Annex 19 of the EIAO-TM, Guidelines for Assessment of Impact On Sites of Cultural Heritage and Other Impacts;
- Guidance Notes on Assessment of Impact on Sites of Cultural Heritage in Environmental Impact Assessment Studies (GN-CH); and
- Hong Kong Planning Standards and Guidelines (HKPSG);
- The list of 1444 historic buildings proposed by Antiquities Advisory Board;
- Guidelines for Cultural Heritage Impact Assessment issued by the Antiquities and Monuments Office (GCHIA).

13.2.2 Antiquities and Monuments Ordinance

13.2.2.1 Legislation relating to antiquities is set out in the Antiquities and Monuments Ordinance (Chapter 53 of the Laws of Hong Kong), which came into force on 1 January 1976. The Antiquities and Monuments Ordinance provides statutory protection against the threat of development on Declared Monuments to enable their preservation for posterity.

13.2.2.2 The Ordinance contains the statutory procedures for the Declaration of Monuments. The legislation applies equally to sites on land and underwater. The purpose of the Ordinance is to prescribe controls for the discovery and protection of antiquities in Hong Kong. A summary of the key aspects of the legislation relevant to the current study is presented below.

13.2.2.3 Human artefacts, relics and built structures may be gazetted and protected as monuments. The Antiquities Authority may, after consultation with the Antiquities Advisory Board (AAB) and with the Chief Executive's approval, declare any place,

building, site or structure which the Antiquities Authority considers to be of public interest by reason of its historical, archaeological or palaeontological significance.

13.2.2.4 The discovery of an Antiquity, as defined in the Ordinance must be reported to the Antiquities Authority (the Authority), or a designated person. The Ordinance also provides that, the ownership of every relic discovered in Hong Kong after the commencement of this Ordinance shall vest in the Government from the moment of discovery. Any person who has discovered an antiquity or suspected antiquity should take all reasonable measures to protect it making reference to Section 11 of Antiquities and Monuments Ordinance.

13.2.2.5 No person, other than the Authority and a designate person authorized by him, shall excavate or search for antiquities except in accordance with a licence granted to him.

13.2.2.6 A licence may not be granted to a person unless the Authority is satisfied that the applicant 1) has had sufficient scientific training or experience to enable him to carry out the excavation and search satisfactorily; 2) has at his disposal sufficient and financial or other resources to enable him to carry out the excavation and search satisfactorily; 3) is able to conduct, or arrange for, a proper scientific study of any antiquities discovered as a result of the excavation and search.

13.2.2.7 Once declared a site of public interest, no person may undertake acts which are prohibited under the Ordinance, such as to demolish or carry on building or other works, unless a permit is obtained from the Antiquities Authority.

13.2.2.8 The Ordinance defines an antiquity as a relic (a moveable object made before 1800) and a place, building, site or structure erected, formed or built by human agency before the year 1800.

13.2.3 Annex 10 of the EIAO-TM, Criteria for Evaluating Visual and Landscape Impact, and Impact on Sites of Cultural Heritage

13.2.3.1 This technical memorandum is issued under section 16 of the EIAO. Annex 10 of the EIAO-TM lists out the criteria for evaluating impact on sites of cultural heritage including:

- The general presumption in favour of the protection and conservation of all sites of cultural heritage because they provide an essential, finite and irreplaceable link between the past and the future and are points of reference and identity for culture and tradition.
- Adverse impacts on sites of cultural heritage shall be kept to the absolute minimum.

13.2.4 Annex 19 of the EIAO-TM, Guidelines for Assessment of Impact On Sites of Cultural Heritage and Other Impacts

13.2.4.1 Annex 19 of the EIAO-TM describes the commonly adopted approaches and methodologies for assessment of impact on sites of cultural heritage. Though there is no quantitative standard in deciding the relative importance of these sites, sites of unique archaeological, historical or architectural value will be considered as highly significant in general. Baseline study, methodology, impact assessment and

mitigation measures form a basic skeleton in determination of impact on sites of cultural heritage.

13.2.5 Guidance Notes on Assessment of Impact on Sites of Cultural Heritage in Environmental Impact Assessment Studies

13.2.5.1 The purpose of this set of guidance notes is to assist the understanding of the requirements set out in section 2 of Annex 10 and Annex 19 of the EIAO-TM in assessing impact on sites of cultural heritage in EIA studies.

13.2.5.2 Under section 9 of the TM, the Director of Environmental Protection shall take advice from the relevant authorities on matters prescribed in that section in considering applications for approval of EIA reports. The relevant authority on antiquities and monuments is the AMO under the Secretary for Home Affairs (previously under the Secretary for Broadcasting, Culture and Sport).

13.2.5.3 This set of guidance notes shall in no way override the provisions under section 2 of Annex 10 and Annex 19 of the EIAO-TM which stipulate the requirements in carrying out assessment of impact on sites of cultural heritage in EIA reports. This set of guidance notes serves only as a reference and is not meant to be exhaustive nor comprehensive. The onus remains on the Project proponents to conduct EIA studies, when required, in such manner and to such standard as to completely satisfy the Director in accordance with the EIAO-TM.

13.2.5.4 This set of guidance notes will be revised from time to time, as appropriate, in light of further experience gained.

13.2.6 Hong Kong Planning Standards and Guidelines

13.2.6.1 The HKPSG is a Government manual of criteria for determining the scale, location and site requirements of various land uses and facilities. As planning standards and guidelines could affect the allocation of scarce land and financial resources, they should be applied with a degree of flexibility. Trade-offs may be necessary so that the community at large could benefit most from the development. Environmental factors and criteria have to be incorporated into the land use planning process in order to prevent adverse environmental problems. Guidance for environmental planning of both public and private developments was given in these guidelines and the guidelines may be applied at three broad planning levels: strategic/territorial, sub-regional and district/local planning.

13.2.7 The list of 1444 historic buildings proposed by Antiquities Advisory Board

13.2.7.1 The AAB announced in March 2009 the completion of the assessment of 1444 historic buildings. Since then, the AAB has been working on confirming the grading of the historic buildings with reference to information from the AMO and opinions received from the public. The AAB will also consider the heritage value and grading of the new items proposed by the public for assessment. In view of the variety of the new items and the tremendous efforts required for research such as document

checking, site-inspection, information recording, etc, the AAB will handle the new items in phases based on the cogent need for assessment.

13.2.8 Guidelines for Cultural Heritage Impact Assessment issued by the Antiquities and Monuments Office

13.2.8.1 This guideline is also appended in the EIA Study Brief of this Study (EIA Study Brief No. ESB-246/2012). The purpose of the guidelines is to assist the understanding of the requirements in assessing impact on archaeological and built heritage. The guidelines which will be revised by AMO from time to time, where appropriate, and when required should be followed in the interest of professional practice.

13.2.8.2 According to the above guideline, a comprehensive CHIA should include a baseline study, an impact assessment study associated with the appropriate mitigation measures proposed and to be implemented by project proponents.

13.3 Cultural Heritage Assessment Area

13.3.1.1 As mentioned in **Section 1.3**, the YLS PDA covers approximately 223.5 hectares (ha) and is located to the south of Yuen Long New Town and bounded by YLH, Kung Um Road and Tai Lam Country Park. The PDA is divided into three areas: the area to the western side that includes TYST (Area 1), the northern part of the land that lies on both sides of Kung Um Road and Kiu Hing Road (Area 2), and the southern part of the land that lies on both sides of Kung Um Road and Pak Sha Shan Road (Area 3). Other than development within the PDA, there are a number of supporting infrastructure works outside the PDA boundaries, which include the following

- New slip roads connecting the PDA to Pok Oi Interchange and a new PTI next to Yoho Midtown;
- New PTI next to Yoho Midtown;
- New TYST Reclaimed Water Service Reservoir and its associated works;
- Partial decking of Yuen Long Nullah along Kung Um Road and Kiu Hing Road;
- Modification works on the trunk roads connecting Tin Shui Wai West (TSWW) Interchange;
- Modification works on the primary distributor roads connecting TYST Interchange;
- Other new local roads connecting PDA;
- Rising Main for water supplies and sewage; and
- New sewer from YLS STW to existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to Urmston Road submarine outfall.

13.3.1.2 All construction works (including site clearance and site formation works) will only be carried out within the PDA boundary and the works boundary outside PDAs. Nevertheless, an assessment area of 100m extending from the PDA boundary and the works boundary outside PDAs (see **Figure 13.1**) will be adopted for the baseline

conditions review and identification of Declared Monuments, Graded Historic Buildings and Sites of Archaeological Interests (SAIs) in the subsequent sections.

13.4 Baseline Conditions

13.4.1 Terrestrial Archaeology and Built Heritage

Geology and Topography

13.4.1.1 The PDA is located at YLS, as shown in **Figure 13.1**. The current locations and extents of the PDA cover about 223.5 hectares. Other supporting infrastructure works are located outside the PDAs and all construction works are confined within the works boundary outside PDAs. No deep excavation is anticipated for the construction of supporting infrastructure works.

13.4.1.2 The physical setting which characterises YLS is one of low bedrock rises which form promontories and outlying hillocks surrounded by wide alluvial valleys. The bedrock which forms the basement geology includes fine grained granite with areas of meta-siltstones toward the north of the PDA (**Figure 13.2**). Colluvium and debris flow deposits surround the low hills, with wide valleys of terraced alluvium. Narrow deposits of recent alluvium are confined to the river and stream courses. The PDA is bounded to the south by prominent granitic bedrock uplands.

13.4.1.3 The PDA boundary and works area outside boundary are located at the northern boundary of the bedrock uplands within the central New Territories – areas where there have been significant landscape change, particularly retreat of sea levels over the past 1000 years. Some 6000 years ago, the former coastline reached the northern section of the PDA and at this time, those areas which are presently wide alluvial valleys would have been shallow seas fringed with marshland and mangrove (Fyfe, Campbell & Lai 2000).

13.4.1.4 Within this environment the low bedrock rises and hillocks formed promontories and islands with close proximity to the sea and marshland resources. Within the PDA, these small hillocks exist within the centre of Tong Yan San Tsuen (唐人新村) (TYST) and within a few areas not currently occupied by urban/industrial development west of Kung Um Road and near Pak Sha Tsuen (白沙村) (**Figure 13.2**).

Historical Background

13.4.1.5 1 historic village (TYST) is located within the PDA, 5 historic villages including Sham Chung Tsuen, Tin Liu Tsuen, Shan Ha Tsuen, Lam Hau Tsuen and Ha Tsuen Shi are located within the 100m assessment area and 10 historic villages are located outside the 100m assessment area. These include villages within the jurisdiction of Shap Pat Heung (十八鄉) and villages within the jurisdiction of Ping Shan (屏山) (**Figure 13.3**). The origins and historic details for these villages have been provided

by Professor Anthony Siu based on consultation with village representatives and are included in **Table 13.1**. A summary of the origins of these villages is provided below.

13.4.1.6 Historic villages which date from Early Song Dynasty through Ming and mid Qing Dynasty provide evidence of occupation during the early Song to mid Qing historic period are located in the immediate surrounds of the PDA.

13.4.1.7 TYST was established in 1932 by Tong Hung-ki and the first residence included the Lams and Tongs from Zhonghsan, Guangdong. Fui Sha Wai (灰沙圍) and Shan Ha Tsuen (山廈村), where both lie outside but in the vicinity of the PDA (near TYST), have origins of the Southern Song Dynasty and Ming Dynasty respectively (Prof Siu p.comm). Proximity of villages of these early dates suggests the prospect of Song period artefacts in and around TYST. In addition, Lam Hau Tsuen (欖口村), which is also outside but in the vicinity of the PDA, was settled from Shan Ha Tsuen (山廈村) in the mid Qing Dynasty.

13.4.1.8 Within the eastern part of the PDA centred around Lam Tai East Road and Kung Um (公庵) Roads, there are no villages with Song Dynasty origins. The earliest settlement of Yuan Dynasty date (1280-1368) at Yeung Ka Tsuen outside the PDA in the far south. Sham Chung Tsuen (深涌村) in the far north of the PDA is also of Yuan Dynasty date. From north to south, the villages of Tin Liu Tsuen (田寮村), Muk Kiu Tau Tsuen (木橋頭村), Pak Sha Tsuen (白沙村) and Wong Nai Tun Tsuen (黃泥墩村) have origins in the late to middle Ming Dynasty with Tai Tong Tsuen (大棠村) of early Qing date.

13.4.1.9 While further away at about 500m to 600m from the PDA boundary, the existing villages of Shui Tsui San Tsuen (水蕉新村) and Shui Tsui Lo Wai (水蕉老圍) have a number of historic buildings and settlement history dates to the late Ming Dynasty.

13.4.1.10 Historic villages associated with sewer works to the north-west of the PDA include Ha Tsuen Shi which contains a Tang Ancestral Hall, Kwan Tai Temple and Gate Tower. Road widening and nullah decking north of the PDA at Lung Tin Tsuen includes two heritage buildings and a study hall.

Table 13.1 Existing villages in the vicinity of the PDA

No.	Village	Surnames	Origin	Arrival	Historic Buildings
Within PDA					
2	TYST (唐人新村)	Tong (唐) and Lam (林)	Zhongshan (中山)	1936	1. Kam Lan Koon(金蘭觀) 2. Tang's grave 3. Yeung's grave
Outside PDA					
1	Fui Sha Wai (灰沙圍)	Tang (鄧)	Jishui, Jiangxi (江西吉水)	South Song dynasty (1216)	1. Entrance Gate
3,4	Lam Hau Tsuen (Lam Hau Wai and Lam Hau Tsuen) 欖口村	Cheung (張)	Dongguan (東莞)（moved from Shan Ha Tsuen）	Mid-Qing dynasty	1. Entrance Gate 2. Shrine
		Man (文), Yip (葉), Wong (黃) and Lok (駱)	-	-	3. Yan Shau Tong (仁壽堂)for worshipping Kwan Tai
5	Shan Ha Tsuen (山廈村)	Cheung (張)	Dongguan (東莞)	South Song period (1216)	1. Entrance Gate 2. Shrine for worshipping Monk Kung Hum 3. Cheung Ancestral Hall (張氏宗祠)
6	Pak Sha Tsuen (白沙村)	Chow (周), Lau(劉), and Tam (譚)	Dongguan(東莞)	Ming dynasty (1368－1644)	1. Entrance Gate 2. Shrine 3. God of Earth
		Yick (易)	Heshan (鶴山)		4. Watch-tower in ruin
		Lee (李) and Yeung (楊)	Siyi (四邑)		5. Ng Fui Study Hall (五奎書室)
		Tao (陶)	-	-	6. Yeungs' Garden (楊苑) with wall and watch-tower inside
7	Tai Tong Tsuen (大棠村)	Leung (梁)	Lisonglang, Dongguan (東莞李松蓢)	Early Qing dynasty (1700s)	
		Lee (李)	Xiamaxiang, Dongguan (東莞下馬鄉)	Early Qing dynasty (1700s)	

No.	Village	Surnames	Origin	Arrival	Historic Buildings
8	Sham Chung Tsuen (深涌村)	Chan (陳)	Nantou, Xinan (新（寶）安南頭)	Later Yuan dynasty(1356)	1. Kung So for worshipping hall
		Yip (葉), Kwan (關) and Lam (林)	-	-	-
9	Tin Liu Tsuen (田寮村)	Wu (胡), Wong (黃), Yip (葉) and Sit (薛)	-	-	-
10	Muk Kiu Tau Tsuen (木橋頭村)	Wu (胡) and Chan (陳)	Sadaonzhu, Zengcheng (增城三道竹)	Mid-Ming dynasty (before 1456)	1. Entrance Gate
11	Shui Chiu San Tsuen (水蕉新村)	Kan (簡), Cheung (張), Wong (黃), Lam (林), Yeung (楊), Cheng (程) and Chen (陳)	Separated from Shui Chiu Lo Wa		1. Chun Fong Cheng Kung Chi (俊芳程公祠)
					2. Chin Fung Lam Kung Chi (振鳳林公祠)
					3. Lam Ancestral Hall (林氏宗祠)
					4. Fook Hing Tong(福慶堂), No.121
					5. Fook Wah Study Hall (福華書室)
					6. Chun Wah Study Hall (俊華書室)
12	Wong Nai Tun Tsuen (黃泥墩村)	Ho (何) and Yu (俞)	Boluo (博羅)	Later Ming dynasty (1500s)	1. Village House no. 12－14
13	Yeung Ka Tsuen (楊家村)	Yeung (楊)	Douchong, Dongguan (東莞斗涌)	Yuan dynasty (1280－1368)	Sik House (適廬), No. 132 that was built in 1930 with two Hakka houses.
14	Shui Chiu Lo Wai (水蕉老圍)	Kan (簡)	Jinzhoudong, Boluo (博羅金州洞)	Later Ming dynasty (about 1640）	1. God of Earth
					2. Kwan Tai Kung (關帝宮)
					3. Kwok Man Study Hall (覺民書室)
					4. Cheung Ancestral Hall (張氏宗祠)
					5. Yeung Ancestral Hall (楊氏宗祠)

No.	Village	Surnames	Origin	Arrival	Historic Buildings
		Tse (謝), Chen (陳), Cheung (張), Lam (林), Yeung (楊), Cheng (程) and Wong (黃)	-	-	6. Cheung Ancestral Hall (張氏宗祠) with two halls and roof type of Ngang Shan
15	Ha Tsuen Shi	Tang (鄧)	-	Ming Dynasty	Gate Tower
					Hwai Tai Temple
					Tang Ancestral Hall
16	Lung Tin Tsuen	-	Guangdong (廣東)	1930s	Lung Tin Study Hal
					Village House 21-22

Previous Archaeological Work

13.4.1.11 There have been two regional surveys conducted in the Yuen Long Area between 1982 and 1997, including:

- Survey done by Peacock and Nixon in 1982 - 1985; and
- Territory-wide archaeological survey (Yuen Long)

13.4.1.12 An archaeological survey was conducted for the EIA and TIA studies for Stage 2 of the Yuen Long and Kam Tin Sewerage and Sewerage Disposal Final EIA report. 10 auger holes and one test pit were excavated to the immediate west of Shan Ha Tsuen. No archaeological remains were found within the auger holes and the test pit revealed undiagnostic village-ware and porcelain within the upper layers. Shallow water table has constrained the investigation in this area.

13.4.1.13 On the other hand, an archaeological monitoring was conducted for the sewer installation works in TYST in 2001. The archaeological monitoring included excavation of 7 test pits. No archaeological remains were identified with the exception of a small unstratified Song Dynasty sherd near test pit 1.

13.4.1.14 An archaeological survey was conducted for HSK EIA. Field survey was conducted in 2015 and included 46 survey locations. 6 auger holes and 1 uncounted hole were drilled to the south side of Hung Uk Tsuen. No archaeological remains were identified, however, due to the location's low hill landscape settings 3 Archaeological Potential Areas (APAs) were circled, while one of them – Hung Uk Tsuen (South) – is situated within the 100m assessment area but outside the PDA.

13.4.1.15 An archaeological survey was conducted as part of the current EIA in 2015 and is elaborated in **Section 13.6**. The survey was conducted on government land on hillcrest areas within TYST Subsurface survey, including a programme of 4 test pit excavations and 11 auger hole drillings, were conducted from 25 June to 8 July 2015. Stratigraphy within all test pits confirmed shallow fill material above shallow soils and debris flow material. Modern debris and fragments of village-ware pottery and undiagnostic provincial porcelain were recovered within all test pits, with Test Pit C containing some Song/Qing dynasty cloth impressed tile fragments – albeit in a disturbed context. This material represented the only objects of potential archaeological significance which suggest early Song/Qing occupation in the vicinity.

13.4.1.16 These archaeological works present data from successive archaeological surveys across the Yuen Long region. **Figure 13.4** shows location of previous studies for terrestrial archaeology. A summary of these archaeological surveys and other relevant sources of information are given below in **Table 13.2**.

Table 13.2 Relevant Previous Studies for Terrestrial Archaeology

Report	Relevance to this Study	Results
Report of the Hong Kong Archaeological Survey, volumes I-III	This report presents the results of the first territory-wide archaeological survey in Hong Kong which included field visits and field evaluation of known and potential sites of	No archaeological sites within the PDA. 5 locations with archaeological potential in adjacent area but outside the 100m assessment area are mentioned, they are: Tung Tau Tsuen (0605), Ha Tsuen (W) (0621), Tseung Kong Wai (0622), San Wai (0623) and Ping Shan

Report	Relevance to this Study	Results
	archaeological interest in the Yuen Long region.	(B) (0619). Among them, only Tung Tau Tsuen has been conducted with site visit and surface observation, the rest 4 locations were not surveyed. None of the locations is provided with marked survey boundary on map.
Territory-wide archaeological survey (Yuen Long) 全港文物普查 1997 第一地區(元朗區)工作報告	As part of the second territory-wide archaeological survey of Hong Kong, Mr. Au Ka–fat led a team and conducted a program of field evaluation throughout the Yuen Long region during 1997-8.	No archaeological sites within the PDA. 9 locations are mentioned, they are: Tseung Kong Wai, Tai Tong Tsuen, Ping Shan Tsuen, Hung Uk Tsuen, Tan Kwai Tsuen, Tang Yan San Tsuen, San Sang Tsuen, Tung Tau Tsuen (close to Ha Tsuen), Tung Tau Tsuen (close to Yuen Long Old Market). Only Tang Yan San Tsuen lies within the PDA, and Tai Tong Tsuen lies close to the 100m assessment area; however, both locations yield no findings. The rest are all outside of the 100m assessment area. Among all 9 locations, only Tseung Kong Wai has 7 test pits trenched, the rest have been conducted with field visits only and no marked surface scanning boundaries were published in the report.
EIA and TIA report for Stage 2 of the Yuen Long and Kam Tin Sewerage and Sewerage Disposal Final EIA report	Part of this EIA included archaeological investigation at Shan Ha Tsuen.	No archaeological sites within the PDA. 2 locations have been surveyed, they are Shan Ha Tsuen and Shung Ching San Tsuen. Both locations have been conducted with 1 test pit and 10 auger holes. Only Shan Ha Tsuen lies within the PDA, Shung Ching San Tsuen's survey location lies outside the 100m assessment boundary.
Archaeological monitoring at Tong Yan San Tsuen Archaeo-Environments Ltd.	Archaeological monitoring conducted for the sewer installation works in TYST within the YLS PDA in 2001	7 test pits were excavated with discovery of one small unstratified Song sherd as a surface find near TP1.
HSK EIA	Part of this EIA included archaeological investigation at Hung Uk Tsuen.	No archaeological sites within the PDA. 6 auger holes and 1 uncounted auger hole have been drilled to south part of Hung Uk Tsuen. Hung Uk Tsuen (South) is situated within the 100m assessment area but outside the PDA.

Terrestrial Archaeological Potential

13.4.1.17 It is considered unlikely that sites or features of cultural heritage beyond the PDA boundary and works boundary outside PDAs will be impacted by project

development works. For this reason the following summary of cultural heritage potential is confined to the PDA boundary and works boundary outside PDAs.

13.4.1.18 There are areas of archaeological potential within the PDA boundary and works boundary outside PDAs, including small hillocks and lower slopes which represent areas of potential prehistoric and early historic period occupation and the edge of the former coastline at this time (see **Figure 13.6**).

13.4.1.19 Some 6000 years ago, the former coastline reached the northern section of the PDA and at this time, those areas which are presently wide alluvial valleys would have been shallow seas fringed with marshland and mangrove (see **Figures 13.5** and **13.6**). Within this environment the low bedrock rises and hillocks formed promontories and islands with close proximity to the sea and marshland resources. These landforms within the PDA would have potential for prehistoric deposits. Within the PDA these small hillocks exist within the centre of TYST and within a few areas not currently occupied by urban/industrial development west of Kung Um Road and near Pak Sha Tsuen. These areas have archaeological potential as sites of prehistoric and historic period occupation (Fyfe, Campbell. and Lai 2000).

13.4.1.20 Prehistoric occupation was recorded at Yuen Long and further to the north-east at Ho Sheung Heung. Prehistoric archaeological material has also been found on lower hill slopes in the interior of the territory including Bronze Age (1500–221 BC) sites at Ngau Tam Mei, Yuen Long (Au 1997). Earlier material from the Late Neolithic (2500-1500 BC) – Bronze Age (1500-221 BC) has been reported from Sheung Shui Wa Shan, Sheung Shui and Hung Leng, in Fanling.

13.4.1.21 In prehistoric times the northern part of the PDA - areas presently reclaimed land, fishponds and river flood plains were shallow seas fringed with marshland and mangrove. Within this environment present hillocks formed islands and islets; therefore the lower hill slopes and former shorelines of these islands may have potential for prehistoric deposits.

13.4.1.22 Within the PDA, there are small hillocks in the centre of TYST and within a few areas not currently occupied by urban/industrial development west of Kung Um Road and near Pak Sha Tsuen (**Figure 13.6**), in and around Sha Ha Tsuen, TYST and north of Wong Nai Tun Tsuen. These areas have moderate archaeological potential as sites of prehistoric or early historic period occupation at the edge of or close proximity to the former shallow sea.

13.4.1.23 Large swathes of the PDA are occupied for industrial development, open storage, residential and village uses as shown in **Figure 13.7**. Prominent areas of residential development, including TYST, are occupied by the people in close association with many factories in the PDA. There are also large areas of unplanned industrial development and open storage in the eastern and south-eastern part of the PDA. These areas will have suffered from heavy ground disturbance and have little or no archaeological potential. In addition, there are highway and infrastructure works constructed since the 1970s, like those in east of TYST and along Kung Um Road, which indicate ground disturbance. The presence of these developments across a large proportion of the PDA will constrain archaeological survey in these areas.

Declared Monuments, Graded Historic Buildings, Sites of Archaeological Interest

Declared Monuments

13.4.1.24 There are two declared monuments within the 100m assessment area summarised in **Table 13.3** below and are shown in **Figure 13.4**. Details of these declared monuments are presented in **Appendix 13.2**.

Table 13.3 Declared Monuments within 100m Assessment Area

No.	Ref. No.	Name	Remarks	Description
1	68	Cheung Ancestral Hall, Shan Ha Tsuen (元朗山廈村張氏宗祠)	Within 100m assessment area but outside PDA)	The Cheung Ancestral Hall was built in 1815 by the Cheung Clan. Placing great importance on early education, the Cheung clan built a number of study halls in Shan Ha Tsuen. The Cheung Ancestral Hall is a traditional two-hall grey brick structure with two side chambers. An altar at the rear hall features two tablets: the soul tablet of Cheung Gau-ling, a prime minister of the Tang dynasty and Dongguan, a pioneer ancestor. The altar is elaborately decorated with fine wood carvings of "The Eight Immortals". Major restoration of the Cheung Ancestral Hall was carried out in 1999 and was under the supervision of the Architectural Services Department and the Antiquities and Monuments Office. The Cheung Ancestral Hall was declared a monument on 30 December 1999.
5	83	Tang Ancestral Hall, Ha Tsuen (元朗廈村鄧氏宗祠)	Within 100m assessment area but outside PDA)	The Tang Ancestral Hall, the Guesthouse and Yau Kung School form a unique historic building compound, representing both the clan history and the development of education in the area. The Tang Ancestral Hall, also known as Yau Kung Tong, was constructed by the Tang clan of Ha Tsuen to commemorate their two founding ancestors, Tang Hung-chi and Tang Hung-wai, for establishing the village settlements in Ha Tsuen. According to a stone inscription placed in the middle hall, construction of the Tang Ancestral Hall began in 1749 and was completed in 1750. The ancestral hall came into use in 1751. The hall underwent two major renovations, one in the 17th year of the Daoguang reign (1837) and the other in the 9th year of the Guangxu reign (1883). Most of the historic building's fabric and relics are still preserved in the hall. There are a number of honorary plaques hanging in the middle hall, indicating the glorious history of the Tangs in the Qing imperial government. The Tang Ancestral Hall is one of the few surviving local examples of Qing vernacular architecture in the three-hall, two-courtyard layout. The historic architectural components and relics are well preserved. Nowadays, the Tang Ancestral Hall still serves as a venue for important religious and ceremonial events of the Tang clan in Ha Tsuen. The Guesthouse and the Yau Kung School adjacent to the Tang Ancestral Hall are believed to have been built before 1924. The Guesthouse was originally built to provide accommodation for guests of the clan. An altar to Man Cheong (God of Literature) and Mo Kuk (God of Martial Arts) is still retained for sacrificial offerings. Yau Kung School was built to provide education for children in the area. Owing to an increase in the number of students, the Guesthouse and part of the ancestral hall were once used as classrooms and as dormitories for teachers.

Source: AMO's Declared Monuments in Hong Kong (as at May 2016)

Graded Historic Buildings

13.4.1.25 There is one graded historic building within PDA identified as shown in **Table 13.4** and **Figure 13.4**. Details of this graded historic building are presented in **Appendix 13.2**.

Table 13.4 Graded Historic Buildings within the PDA

No.	Ref. No. [1]	Name and Address[1]	Confirmed Grading[1]	Ownership [1]	Remarks	Description
1	1045	Yeung Hau Temple, TYST, Ping Shan, Yuen Long, N.T. 新界元朗屏山唐人新村楊侯古廟	3	Private	Within PDA	The Temple in TYST is one of the six Yeung Hau temples in Yuen Long. It was built in 1711 in the Qing dynasty. TYST was established in 1932 and settled by the Lams (林) and Tongs (唐) from Zhongshan county (中山) of Guangdong (廣東). After the WWII, it was inhabited by Shanghai dialect speaking immigrants from the mainland and by whom the temple was not supported. Deities Lady Kam Fa (金花夫人) and Nam Hoi Ku Fat (南海古佛) are also worshipped in side bays of the temple. The temple is a vernacular building of a two-hall-one-courtyard plan having a three bays layout. It is constructed of green bricks with walls supporting pitched roofs. Above the recessed entrance is the name board of the temple. Wall friezes are decorated with paintings and motifs. Its front façade is with plastered mouldings. Gable friezes are with curling grass pattern. Its fascia board at the entrance is with carving decoration. Nowadays, the custom of *diandeng* (點燈) is still conducted at the temple for the birth of new male babies. (Reference taken from: http://www.aab.gov.hk/historicbuilding/en/1045_Appraisal_En.pdf)

Note:

[1] As per AMO's List of the Historic Buildings in Building Assessment (as of 8 June 2017)

13.4.1.26 Apart from the Yeung Hau Temple, which is a graded historic building within the PDA as identified and described above, there are altogether 14 graded historic buildings, 1 proposed graded historic building and 1 nil grade historic building located within 100m Assessment Area but outside PDA. The existing village at Shan Ha Tsuen contain most of these buildings. A summary of the built heritage within 100m assessment area is shown in **Table 13.5** below and details are presented in **Figure 13.4** and **Appendix 13.2**.

Table 13.5 Built heritage within 100m Assessment Area

No.	Ref. No.	Name and Address	Confirmed Grading	Ownership	Remarks
1	315	Gate Tower, Ha Tsuen Shi (廈村市門樓)	2	Private	within 100m Assessment Area but outside PDA
2	541	Tat Yan Study Hall, No. 226 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 226 號達仁書室)	2	Private	
3	544	No. 329 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 329 號)	2	Private	
4	545	No. 330 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 330 號)	2	Private	
5	546	No. 331 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 331 號)	2	Private	
6	547	No. 332 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 332 號)	2	Private	
7	549	Kwan Tai Temple, Ha Tsuen Shi (廈村市關帝廟)	2	Private	
8	741	Entrance Gate, Lam Hau Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山欖口村圍門)	3	Private	
9	771	Shrine, Lam Hau Tsuen, Ping Shan,Yuen Long, N.T. (新界元朗屏山欖口村神廳)	3	Private	
10	819	Ng Fui Study Hall, No. 116 Pak Sha Tsuen, Shap Pat Heung, Yuen Long, N.T. (新界元朗十八鄉白沙村 116 號五奎書室)	3	Private	
11	942	Entrance Gate, No. 7 Tin Liu Tsuen, Shap Pat Heung, Yuen Long, N.T. (新界元朗十八鄉田寮村 7 號圍門)	3	Private	
12	971	Main Shrine, No.73 Tin Liu Tsuen, Shap Pat Heung, Yuen Long, N.T. (新界元朗十八鄉田寮村 73 號神廳)	3	Private	
13	1011	Hing Po Study Hall, No. 162 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 162 號興寶書室)	3	Private	
14	1043	Yan Shau Tong, No. 150 Lam Hau Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山欖口村 150 號仁壽堂)	3	Private	
15	893	Village House 23, Lung Tin Tsuen (龍田村 23 號)	Proposed Grade 3	Private	
16	1200	Lung Yip Yuen, No. 198 Shan Ha Tsuen, Ping Shan, Yuen Long, N.T. (新界元朗屏山山下村 198 號農業園)	Nil grade	Private	

Source: AMO's List of the 1444 Historic Buildings in Building Assessment (as at 8 June 2017)

Sites of Archaeological Interest

13.4.1.27 AMO maintains a list of Sites of Archaeological Interest which is updated from time to time. This list can be consulted at the AMO, or from other approved EIA reports in EPD's EIAO Register Office. However, the list is neither meant to be exhaustive, nor is the information contained therein comprehensive, particularly in the case of sites of archaeological interest or cultural features buried underground. Other useful sources of relevant information include the tertiary institutions (e.g. the Hong Kong Collection at the University of Hong Kong Library, Departments of History and Architecture at the University of Hong Kong and the Chinese University of Hong

Kong), public libraries and archives (e.g. Public Records Office), District Offices, District Lands Offices and Land Registries, etc.

13.4.1.28 According to the available information, Tseung Kong Wai is the only Site of Archaeological Interest within the 100m assessment area but outside the PDA. Other two Sites of Archaeological Interest are the Sheung Cheung Wai (上章圍) Site of Archaeological Interest, at approximately 800m, and the Yuen Leng (圓嶺) Site of Archaeological Interest locates, at 1500m, from the PDA respectively.

Graves

13.4.1.29 A surface inspection of graves in the PDA as part of the baseline assessment has previously been conducted and 33 graves were identified. However, none of these graves yielded dates older than 100 years (no earlier than 1930). While these graves represent the ancestral graves of the local villages and therefore provide important information about places of cultural importance and intangible heritage, they would appear limited for reconstructing earlier periods of settlement and archaeological potential. Access limitations and safety concerns prevented further on-site investigation.

13.4.1.30 A summary of information for graves within 100m assessment area is listed in **Table 13.6** below and their locations are shown in **Figure 13.8**.

Table 13.6 Summary of Graves and Land Use within 100m Assessment Area

Grave No.	Location	Land Use in RODP
1	TYST	Storage (OU)
2	TYST	Storage (OU)
3	TYST	Storage (OU)
4	TYST	Storage (OU)
5	TYST	Storage (OU)
6	TYST	Amenity Area (A)
7	TYST	Amenity Area (A)
8	TYST	Amenity Area (A)
9	TYST	Amenity Area (A)
10	TYST	Residential (R)
11	TYST	Green Belt (GB)
12	TYST	Residential (R)
13	TYST	Residential (R)
14	TYST	Residential (R)
15	TYST	Outside PDA
16	TYST	Outside PDA
17	Shan Ha Tsuen	Road
18	Between the Lam Tai East Road and Kung Um Road	Residential (R)

Grave No.	Location	Land Use in RODP
19	TYST	Road
20	TYST	Road
21	Pak Sha Tsuen	Road
22	Pak Sha Tsuen	Residential (R)
23	Pak Sha Tsuen	Reedbed
24	Pak Sha Tsuen	Local Open Space (LO)
25	Pak Sha Tsuen	Local Open Space (LO)
26	Pak Sha Tsuen	Local Open Space (LO)
27	Pak Sha Tsuen	Reedbed
28	Pak Sha Tsuen	Reedbed
29	Pak Sha Tsuen	OU (STW)
30	Pak Sha Tsuen	Local Open Space (LO)
31	Wong Nai Tun Tsuen	Outside PDA
32	TYST	Amenity Area (A)
33	TYST	Storage (OU)

13.5 Assessment Methodology

13.5.1 Assessment Scope

13.5.1.1 The assessment of the potential direct and indirect impacts on sites of cultural heritage will be conducted within the PDA boundary and works boundary outside PDA, as shown on **Figure 13.1**.

13.5.1.2 As stipulated in Appendix K-1of the EIA Study Brief no. ESB-246/2012, the CHIA will follow the criteria and guidelines as stated in Annexes 10 and 19 of the EIAO-TM. The key stages for CHIA include the following:

- Baseline study (including a desk-top study and, subject to the result of the desk-top study, a field survey);
- Evaluate impact; and
- Recommend mitigation measures.

13.5.2 Terrestrial Archaeology

Desk-top Review

13.5.2.1 A comprehensive desktop review (as described in **Section 13.3.1**) has been conducted to collate all desktop information available that is relevant to the terrestrial archaeology of YLS.

Field Visit

13.5.2.2 Field inspections were conducted in March and April 2013 to inspect areas of archaeological potential and access. These site visits were conducted to collect and collate existing archaeological information about the PDA and immediate surrounds.

Field Survey Methodology

13.5.2.3 The aim of terrestrial archaeological survey is to obtain field data to assess the archaeological potential within the YLS PDA. The key steps for the terrestrial archaeological field survey are:

- Review available desktop information relevant to terrestrial archaeology including but not limited to historical information, geological, typological and archaeological background as well as previous and present land uses within the PDA;
- Identify any data gap based on available desktop information that may require further data by carrying out the terrestrial archaeological survey;
- Develop an archaeological predictive model which will provide a framework and requirements for a terrestrial archaeological survey plan which includes fieldwork methodology, rationale of quantity and location of auger holes and test pits;

- Carry out the fieldworks in accordance with the terrestrial archaeological survey plan to obtain sufficient field data; and
- Prepare a report containing all the data and results of the terrestrial archaeological survey.

13.5.2.4 The archaeologist responsible for conducting an archaeological fieldwork is required to apply for a Licence to Excavate and Search for Antiquities in accordance with the Antiquities and Monuments Ordinance (Cap.53). The licence shall be in the prescribed form and specify the area of land in respect of which it is granted. It may be granted or renewed for a period not exceeding 12 months. The archaeological survey would collect sufficient information to access the archaeological potential for the purpose of the CHIA as required in Appendix K-1 of EIA Study Brief No. ESB-246/2012. The survey results would be analyzed and combined with the desktop information to establish the baseline information.

13.5.2.5 A Terrestrial Archaeological Survey Proposal (TASP) was submitted to AMO for comment in November 2014. The tentative field program includes 19 test pits and 108 auger holes which focuses on areas near TYST and its surrounding, and West of Muk Kiu Tau Tsuen. As shown in Figure 13.9 which integrates prior landscapes, current land uses and land disturbance, the resulting areas of moderate and low-moderate archaeological potential represent relatively undisturbed areas – commonly elevated areas and hillocks undisturbed by urbanisation or settlement.

13.5.2.6 However, according to the TASP, as access to private lands will constrain the terrestrial archaeological survey, it is proposed that private lands be subject to archaeological survey following land resumption. Therefore, it is proposed to conduct field work in two stages (i.e. Stage 1 for areas of Government land and Stage 2 for private lands).

13.5.2.7 Archaeological survey excavation in Stage 1 of the survey for areas of Government land was conducted on 25 June to 8 July 2015 (Licence No. 387 issued on 15 June 2015). As this survey was conducted as part of the current EIA a summary has been included in the Baseline Section (**Section 13.4**). The survey on Government land has been conducted and results have also been presented in detail in **Section 13.6**. However, as discussed above, other than those existing government lands surveyed, the rest of the lands within the PDA boundary are private lands, which are constrained by accessibility issues and hence are not available for archaeological survey during the EIA programme. Archaeological survey is recommended to be conducted at the private land with moderate archaeological potential instead of on private land with low-moderate archaeological potential based on the findings of archaeological survey conducted in 2015. This is based on discovery of Song/Qing impressed tiles on government land from a disturbed stratigraphy of Test Pit C during this survey. The discovery of Song and Qing pottery in 2015 near the TSWW Interchange and Shan Ha Tsuen, while confined to a relatively small part of the PDA supports the predictive model developed within the TASP, and that areas of "moderate" archaeological potential should be the subject of further archaeological survey on private land after land resumption. Further archaeological survey shall be conducted after land resumption when the constraints on site accessibility are removed before construction commencement. The scope and programme of the proposed archaeological work shall be agreed with AMO prior to the commencement of archaeological survey. Subject to the findings of the archaeological work,

appropriate mitigation measures would be proposed by the Project proponent in prior agreement with the AMO before the construction phase of the proposed development.

13.5.3 Built Heritage

Desktop Review

13.5.3.1 A desktop study has been conducted to reveal information available in the public domain and in accordance with the following legislative standards and guidelines:

- Environmental Impact Assessment Ordinance;
- Antiquities and Monument Ordinance; and
- Hong Kong Planning Standards and Guidelines

13.5.3.2 The public information sources referenced includes the following:

- List of Proposed and Declared Monuments, List of Proposed and Graded Historic Buildings and List of Government Historic Sites as issued by the AMO (however, the list is neither meant to be exhaustive, nor is the information contained therein comprehensive, particularly in the case of archaeological sites or cultural features buried underground);
- Published and unpublished papers and studies, including those kept in tertiary institutions (e.g. the Hong Kong Collection at the University of Hong Kong Library, Departments of History and Architecture at the University of Hong Kong and the Chinese University of Hong Kong), public libraries and archives (e.g. the reference libraries under LCSD, the Public Records Office), District Offices, District Lands Offices and Land Registries, etc.;
- Publications on relevant historical, anthropological and other cultural studies;
- Unpublished archaeological investigation and excavation reports kept by AMO and relevant information from AMO's website;
- Unpublished archival, papers, records; collections and libraries of tertiary institutions;
- Historical documents which can be found in Public Records Office, Lands Registry, District Lands Office, District Office, Museum of History;
- Cartographic and pictorial documentation;
- Existing geological and topographic information; and
- Discussion with local informants.

Field Visit

13.5.3.3 Based on the information of desktop study (see **Section 13.4**), field visits were conducted among the villages within the RODP, to collect and confirm existing built

heritage information within the PDA boundary and works boundary outside PDAs that were not covered by the desktop study. Field visits were conducted in March and April 2013 to inspect areas of built heritage potential, access, existing villages and graves.

Field Survey Methodology

13.5.3.4 For the purpose of this field survey, the scope of built heritage resources has been based on the EIA Study Brief No. ESB-246/2012 Appendix K-1 as follows:

- all declared monuments;
- all proposed monuments;
- all buildings/ structures/ sites graded or proposed to be graded by the AAB;
- Government historic sites identified by AMO; and
- buildings/ structures/ sites of high architectural / historical significance and interest which are not included in items listed above.

13.5.3.5 Cultural landscapes include places associated with historic event, activity, or person or exhibiting other cultural or aesthetic values, such as sacred religious sites, battlefields, a setting for buildings or structures of architectural or archaeological importance, historic field patterns, clan graves, old tracks, fung shui woodlands and ponds, and etc. Each of the villages mentioned above was scanned following assessable foot paths/streets, and where available, local elders or informants have been interviewed to collect information on local history and genealogy.

13.6 Impact Assessment

13.6.1 Terrestrial Archaeology

Summary of Archaeological Potential

13.6.1.1 An archaeological survey (**Appendix 13.1**) was conducted by Archaeological Assessments from June 2015 to July 2015. As this survey was conducted as part of this EIA, a summary of the results have also been added to the Baseline Section (**Section 13.4**). The survey has only assessed for two areas of Government land which is in Area 1 of the PDA, as Areas 2 and 3 are mainly private lands and can only be assessed after land resumption.

13.6.1.2 The two areas of small hillocks were surveyed in turn as the northern and southern study areas respectively (Figure 1 in **Appendix 13.1**). A field scan was conducted for both areas. Disturbance, terracing and refuse was a common feature which affected ground visibility. No archaeological material was recorded. The results of the archaeological survey of the northern area of the surveyed areas of Government land near TYST suggest the lack of archaeology, therefore the archaeological impacts will be low. For the southern part of the surveyed area, the presence of cloth-impressed tiles of Song/Qing Dynasty date suggested some occupation of the area in historic times. Mitigation measure for the southern part is given in **Section 13.7.1**.

13.6.1.3 For works boundary outside PDA, The impacts of the new road alignments in TYST and TSWW Interchange, new sewer and Reclaimed Water Service Reservoir under

the RODP will not require archaeological assessment. These proposed roads and sewer network are located at a highly disturbed and urbanised area, and the proposed Reclaimed Water Service Reservoir is situated on an existing hill slope, it is predicted that the archaeological potential are likely low. These areas will not be surveyed after land resumption. There are some small areas to the west and south of the TYST Interchange which will be surveyed after land resumption.

13.6.1.4 As mentioned in **Section 13.5.2**, due to constraints on site accessibility, no archaeological investigation on private lands could be made. Based on the TASP which included a literature review and development of an archaeological predictive model for the PDA and surrounding region, it is recommended that areas of "moderate" archaeological potential should be subject to survey in two stages, prior to land resumption limited areas of Government land were available and were based on the findings of archaeological survey conducted in 2015 (Archaeological Investigation Report, 2016 as presented in **Appendix 13.1**), given that the archaeological investigation on government land identified only Song/Qing impressed tiles from a disturbed stratigraphy of Test Pit C in the area of moderate archaeological potential. The discovery of Song and Qing pottery near the TSWW Interchange and Shan Ha Tsuen, while confined to a relatively small part of the PDA supports the predictive model developed within the TASP, and that areas of "moderate" archaeological potential should be the subject of further archaeological survey on private land after land resumption. Further archaeological surveys shall be conducted after land resumption when the constraints on site accessibility are removed before construction commencement. The scope and programme of the proposed archaeological work shall be agreed with AMO prior to the commencement of archaeological survey. Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project proponent in prior agreement with the AMO before the construction phase of the proposed development.

Subsurface survey results

13.6.1.5 Subsurface survey, including a programme of 4 test pit excavations and 11 auger hole drillings, were conducted from 25 June to 8 July 2015.

Test pit and auger hole sampling

13.6.1.6 Test pit and auger surveys were conducted in the areas specified in **Appendix 13.1** and the following results were noted:

- The northern area was tested with four auger holes in both upper and lower slope settings. No archaeological material was recorded.
- The southern area was tested with seven auger holes located in upper mid and lower slope settings. No archaeological material was recorded.
- Two test pits (Test Pit A and Test Pit B) (Drawings G.1 in Appendix **13.1**) were excavated in the northern area and two test pits were excavated in the southern area (Test Pit C and Test Pit D) (Drawings G.2 in **Appendix 13.1**). Stratigraphy within all test pits confirmed shallow fill material above shallow soils and debris flow material. Modern debris and fragments of village-ware pottery and undiagnostic provincial porcelain were recovered within all test pits, with Test

Pit C containing some Song/Qing dynasty cloth impressed tile fragments – albeit in a disturbed context. This material was the only objects of potential archaeological significance which suggest early Song/Qing occupation in the vicinity.

13.6.1.7 It was concluded in the archaeological survey that both northern and southern areas were undisturbed until recently and that the northern area has low archaeological potential. For the southern area, the presence of cloth-impressed tiles of Song/Qing Dynasty date suggested some occupation of the area in historic times. The lower slopes of the southern area were proven to contain areas of fill and poor access within private lands.

13.6.1.8 It was a recommendation of the Archaeological Investigation Report as presented in **Appendix 13.1** that an archaeological review should be conducted on availability of more detailed information by the project proponent to assess the archaeological impact on part of the "Southern Study Area" in Area 1 and propose appropriate mitigation measures in consultation with AMO during the next stage of the EIA. A review has been undertaken which indicates that part of the area in question will be developed as Residential Zone 5 (R5) and Local Open Space (LO). This area was identified as occupied by "artificial fill and private lands" and was not accessible during the survey in 2015. While the depth of fill is unknown, private land will become available for survey in land resumption. It is therefore recommended that this area will be the subject of further archaeological survey on land resumption to assess the impacts of development. The area is shown in **Figure 13.10** and is indicative, reflecting the area to be impacted and therefore the area to be surveyed. Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development

13.6.1.9 Predicted impacts within RODP and supporting infrastructures are confined to areas with moderate archaeological potential as shown in **Figure 13.10.** The concerned landuses, possible impact and proposed mitigation measures are summarised in **Table 13.7** below:

Table 13.7 Predicted Impacts within RODP and Supporting Infrastructures

Proposed Landuses	Description of works	Possible Archaeological Potential /Heritage Impacts	Proposed Mitigation Measures
Area near Tin Shui Wai West Interchange (Inset 1)			
TSWW Interchange	Improvement works to existing TSWW Interchange	Impacts around edge of interchange works on area of moderate archaeological potential	Archaeological survey will be conducted after the land resumption before the commencement of the proposed development. The scope and programme of the proposed archaeological work
Road L20	Local distributor with 2 lanes	Impacts across area of moderate archaeological potential	
Road L22	Local distributor with 2 lanes	Impacts across area of moderate archaeological potential	
R5	Low-rise residential development with plot ratio of 1 for private housing	Impacts from residential development across area of moderate archaeological potential	
OU(SU)2	Proposed Multi-Storey Buildings for Storage uses	Impacts from open storage development on area of moderate archaeological potential	

Proposed Landuses	Description of works	Possible Archaeological Potential /Heritage Impacts	Proposed Mitigation Measures
OU(OS)	Area reserved for Open Storage Uses near TSWW interchange	Impacts from open storage development on area of moderate archaeological potential	shall be agreed with AMO. Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project Proponent in prior agreement with the AMO. The proposed survey areas have been reviewed based on the proposed development and current site condition.
A	Slope works and Road Side Amenity	Impacts across area of agricultural land and hillcrest. Area of moderate archaeological potential	
District Open Space	Area designated for Park and recreational uses	Works on district open space on area of moderate archaeological potential	
Area near Tong Yan San Tsuen (Southern study area adjacent location of cloth impressed tiles) (Inset 2)			
R5	Low-rise residential development with plot ratio of 1 for private housing	Impacts from residential development across area of moderate archaeological potential	An archaeological review has recommended an archaeological survey in this area on land resumption Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development.
Road L19	Local distributor with 2 lanes	Impacts across area of moderate archaeological potential	
Local Open Space	Area designated for Park and recreational uses	Works on district open space on area of moderate archaeological potential	
Area near Shan Ha Tsuen (Inset 3)			
R2c	Mid-rise residential development with plot ratio of 4 for private housing	Low impact as it is currently disturbed by open storage	Nil
G	Divisional Fire Station and Ambulance Depot	Low impact as it is currently disturbed by open storage	
RSc	High-rise residential development with plot ratio of 5 for public housing	Impacts from residential development across area of moderate archaeological potential	Archaeological survey will be conducted after the land resumption before the
OU(Hillside River Corridor and Scenic Cycle Track)	A proposed new retention lake and a new water channel along Ma Shan with cycle track and footpath,	Impacts across area of agricultural land and hillcrest. Area of moderate archaeological potential	

Proposed Landuses	Description of works	Possible Archaeological Potential /Heritage Impacts	Proposed Mitigation Measures
A	Slope works and Road Side Amenity	Impacts across area of agricultural land and hillcrest. Area of moderate archaeological potential	commencement of the proposed development. The scope and programme of the proposed archaeological work shall be agreed with AMO. Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project Proponent in prior agreement with the AMO. The proposed survey areas have been reviewed based on the proposed development and current site condition.
Road D1	District Distributor with 4 lanes	Impacts across area of agricultural land and hillcrest. Area of moderate archaeological potential	

Note:
The land use "IC" within Area near Tin Shui Wai West Interchange would be excluded in the impact assessment as there is no development within this land use zoning.
The EFTS will follow the alignment of various new roads, in which the archaeological survey, if required, will be carried out after land resumption of YLS development. Thus there will not be any archaeological impact associated with the site formation of EFTS.

13.6.2 Built Heritage

13.6.2.1 The Yeung Hau temple at TYST is identified as a Grade 3 historic building and should be preserved within the PDA. There are two declared monuments (Cheung Ancestral Hall at Shan Ha Tsuen and Tang Ancestral Hall at Ha Tsuen) within 100m assessment area but outside PDA. Another 14 graded historic buildings, 1 proposed graded historic building and 1 nil grade historic building are located within 100m assessment area but outside PDA. Therefore, impacts on built heritage resources are anticipated to be indirect and of low significance.

13.6.2.2 A total of 33 graves were recorded during the built heritage survey. No dates earlier than 1930 were observed on clan graves. The distribution of clan and individual graves is shown in **Figures 13.8** and summary of grave information is provided in **Table 13.6**. Potential management measures are mentioned in **Section 13.7**.

13.7 Mitigation Measures

13.7.1 Terrestrial Archaeology

13.7.1.1 Predicted impacts within RODP and supporting infrastructures are confined to areas with moderate archaeological potential not covered by the terrestrial archaeological

survey under this Project, and part of the "Southern Study Area" in Area 1 subject to archaeological review and adjacent the location of Song/Qing cloth impressed tiles found in the government land investigated in the terrestrial archaeological survey**.** This review has recommended that this area be subject to archaeological survey prior to construction and that subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. The concerned landuses, possible impact and proposed mitigation measures for areas of moderate archaeological potential are summarised in **Table 13.7.** The extent of archaeological survey required after land resumption is illustrated in **Figure 13.10**, is indicative and is focused on three areas within the PDA and 100m buffer. The areas proposed for archaeological survey represent areas of moderate archaeological potential to be impacted by development across interpreted natural or undisturbed ground. It would be expected that the proposed survey areas would be fine-tuned during the archaeological licencing process, e.g. prior to commencement of archaeological survey on land resumption. For the areas with low-moderate archaeological potential, AMO should be informed immediately in case of discovery of antiquities or supposed antiquities in the course of the construction works. Agreement from AMO would be sought on the follow-up actions if required.

13.7.2 Built Heritage

13.7.2.1 The Yeung Hau temple at TYST is identified as a Grade 3 historic building and has been preserved within the PDA. Impacts on the Yeung Hau temple are likely to be indirect however mitigation is recommended by way of a site audit for noise, dust, visual and vibration effects.

13.7.2.2 The presence of this temple within the YLS PDA offers the opportunity for heritage promotion within the community. Built heritage in the form of temples, ancestral halls and buildings throughout the area also offers the opportunity for incorporation of historic buildings into heritage trails or visitor areas which are subject to further study to be conducted during detailed design stage. Once the idea of heritage trails is pursued, the implementation and management agencies shall also be identified before operation.

13.7.2.3 The planned sewer works to the north-west of the PDA near Ha Tsuen Shi are expected to impact a narrow disturbed footprint along Tin Ha Road south of the village. The impact of these works on the only declared monument, i.e. Tang Ancestral Hall locates 90m away, are expected to be nil.

13.7.2.4 As discussed above a total of 33 graves were recorded within the 100m assessment area (see **Figure 13.8**). **Table 13.8** summarises the potential management measures.

Table 13.8 Summary of graves and corresponding potential mitigation measures

Grave No.	Location	Land Use in RODP	Potential Management Measures
1	TYST	Storage (OU)	Retain
2	TYST	Storage (OU)	Retain
3	TYST	Storage (OU)	Retain

Grave No.	Location	Land Use in RODP	Potential Management Measures
4	TYST	Storage (OU)	Relocate
5	TYST	Storage (OU)	Relocate
6	TYST	Amenity Area (A)	Retain
7	TYST	Amenity Area (A)	Retain
8	TYST	Amenity Area (A)	Retain
9	TYST	Amenity Area (A)	Retain
10	TYST	Residential (R)	Relocate
11	TYST	Green Belt (GB)	Retain
12	TYST	Residential (R)	Relocate
13	TYST	Residential (R)	Relocate
14	TYST	Residential (R)	Relocate
15	TYST	Outside PDA	Retain
16	TYST	Outside PDA	Retain
17	Shan Ha Tsuen	Road	Relocate
18	Between the Lam Tai East Road and Kung Um Road	Residential (R)	Relocate
19	Tong Yan San Tsuen	Road	Relocate
20	Tong Yan San Tsuen	Road	Relocate
21	Pak Sha Tsuen	Road	Relocate
22	Pak Sha Tsuen	Residential (R)	Relocate
23	Pak Sha Tsuen	Reedbed	Retain
24	Pak Sha Tsuen	Local Open Space (LO)	Retain
25	Pak Sha Tsuen	Local Open Space (LO)	Retain
26	Pak Sha Tsuen	Local Open Space (LO)	Retain
27	Pak Sha Tsuen	Reedbed	Retain
28	Pak Sha Tsuen	Reedbed	Retain
29	Pak Sha Tsuen	OU (STW)	Relocate
30	Pak Sha Tsuen	Local Open Space (LO)	Retain
31	Wong Nai Tun Tsuen	Outside PDA	Retain
32	TYST	Amenity Area (A)	Retain
33	TYST	Storage (OU)	Retain

13.8 Residual Impacts

13.8.1.1 Archaeological survey will be conducted at areas of moderate archaeological potential after land resumption prior to commencement of construction. The concerned landuses, possible impact and proposed mitigation measures are summarised in **Table 13.7.** The extent of archaeological survey required after land resumption is illustrated in **Figure 13.10** and is focused on three areas within the PDA and 100m buffer. The areas proposed for archaeological survey represent areas of moderate archaeological potential to be impacted by development across interpreted natural or undisturbed ground.

13.8.1.2 Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project proponent in prior agreement with the AMO before the construction phase of the proposed development. In the previous surveyed government land area, part of the "Southern Study Area" in Area 1 adjacent the location of Song/Qing cloth impressed tiles has been subject to an archaeological review which has recommended that this area be subject to archaeological survey prior to construction. Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. Given the archaeological finds, if any, will be identified and consulted AMO for proper follow-up actions, residual impact is thus not anticipated.

13.9 Conclusions and Recommendations

13.9.1.1 A desk-top study has been conducted to identify cultural heritage resources within the PDA boundary and works boundary outside PDA. There are no Sites of Archaeological Interest within the PDA boundary and works boundary outside PDA. A terrestrial archaeological survey has been conducted on Government land in the TYST area. It was concluded that surveyed areas were undisturbed until recently, and that the part of the area in the north has low archaeological potential. For the private lands that have "Moderate" archaeological potential within the PDA, they are constrained by accessibility issues and hence surveys cannot be conducted during this EIA. In the previous surveyed government land area, part of the "Southern Study Area" in Area 1 adjacent the location of Song/Qing cloth impressed tiles has been subject to an archaeological review which has recommended that this area be subject to archaeological survey prior to construction. Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. Further archaeological survey will be conducted in the area presented in **Table 13.7** after land resumption. The concerned landuses, possible impact and proposed mitigation measures are summarised in **Table 13.7.** The extent of archaeological survey required after land resumption is illustrated in **Figure 13.10** and is focused on three areas within the PDA and 100m buffer. The areas proposed for archaeological survey represent areas of moderate archaeological potential to be impacted by development across interpreted natural or undisturbed ground.

13.9.1.2 The scope and programme of the proposed archaeological work will be agreed with AMO. Subject to the findings of the archaeological work, appropriate mitigation measures would be proposed by the Project proponent in prior agreement with the AMO before the construction phase of the proposed development.

13.9.1.3 The impacts of the new road alignments in TYST and TSWW Interchange, new sewer and Reclaimed Water Service Reservoir under the RODP will not require

archaeological assessment. These proposed roads and sewer network are located at a highly disturbed and urbanised area, and the proposed Reclaimed Water Service Reservoir is situated on an existing hill slope, it is predicted that the archaeological potential are likely low. These areas will not be surveyed after land resumption. There are some small areas to the west and south of the TYST Interchange which will be surveyed after land resumption.

13.9.1.4 As mentioned in **Section 13.6.1**, given the results in southern part of the surveyed Area 1, an archaeological review has recommended that this area be subject to archaeological survey prior to construction. Subject to the findings of archaeological work, appropriate mitigation measures would be proposed by the project proponent in prior agreement with the AMO before the construction phase of the proposed development. (shown in **Figure 13.10**). For the areas with low-moderate archaeological potential, AMO should be informed immediately in case of discovery of antiquities or supposed antiquities in the course of the construction works. Agreement from AMO would be sought on the follow-up actions if required.

13.9.1.5 A Grade 3 historic building, the Yeung Hau temple, lies within the PDA at TYST and should be preserved. There are two declared monuments (Cheung Ancestral Hall at Shan Ha Tsuen and Tang Ancestral Hall at Ha Tsuen) within 100m assessment area but outside PDA. In addition 14 graded, 1 proposed graded and 1 nil graded historic building are identified within 100m assessment area but outside PDA and these buildings will not be impacted by works.

13.10 References

AMO, List of Sites of Archaeological Interest in Hong Kong (as at November 2012).

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